Practice Problems
Calculus Physics II

Name: _____ **Date:** _____

Chapter 50 – Electric Generators, Motors and Power Transmission

1) An electric motor is very similar to

(A) an electric generator.
B) an automobile battery.
C) a radio receiver.
D) none of the above

2) Ignoring the effects of friction, a cyclist will coast farther when a lamp powered by a wheel generator is

A) of low current.
B) of high current.
(C) turned off.
D) none of the above

no counter torque in generator when there is no current.

3) What creates the back emf in an electric motor while it is running?

(A) An induced emf in the windings that oppose the applied potential.
B) An increased resistance in the windings.
C) Air resistance as it turns.
D) The counter torque from the current in the windings.
E) none of the above

4) For an electric motor, how does the current to start the motor running relate to the current to keep the motor going.

A) It takes the same amount of current to start the motor and to keep the motor going.
(B) It takes more current to start the motor than to keep the motor going.
C) It takes less current to start the motor than to keep the motor going.

5) An electric motor is wired to a 120 V rms dc power source, the motor has an internal resistance of 8 Ω and when it is running draws a current from the supply of 1 A, determine the back emf produced by the motor.

$V_{net} = Ir = V_{ext} - V_{emf}$ $V_{emf} = V_{ext} - Ir = 112\,V$

6) An electric motor is wired to a 120 V rms dc power source, the motor has an internal resistance of 8 Ω and when it is running draws a current from the supply of 1 A, determine the current require to start the motor.

$I = \frac{V_{ext}}{r} = 15\,A$ $V_{emf} = 0\,V$

7) An electric car's motor requires 800 V to operate, if the motor has an internal resistance of 0.5 Ω, what current is required to start the engine?

$V_{emf} = 0V$

$I = \frac{V_{ext}}{r} = 1600\,A$

8) An electric car's motor requires 800 V to operate, if the motor has an internal resistance of 0.5 Ω, what current is required to keep the engine running after some time has passed if the back emf generated is 600 V?

$I_r = V_{rms} - V_{Rrms}$ | $I = \frac{V_{rms} - V_{Rrms}}{r} = 400A$

9) A 240 V rms power source is running an electric motor, as the motor is running it create a back emf of 215 V rms and has an internal resistance of 12Ω. Determine current drawn by the motor as it is running.

$I = \frac{V_{rms} - V_{Rrms}}{r} = 2A$

10) A device that transforms mechanical energy into electrical energy is a

A) motor.
B) magnet.
C) transformer.
D) generator.
E) none of these

11) A device that transforms electrical energy to mechanical energy is a

A) generator.
B) motor.
C) transformer.
D) magnet.
E) none of the above

12) The current produced by a common electrical generator is

A) dc.
B) ac.
C) neither of these

13) A generator armature is more difficult to rotate when supplying electric current due to

F) an induced magnetic field in the windings that resists change.
G) increased resistance.
H) energy leaking as heat.
I) none of the above

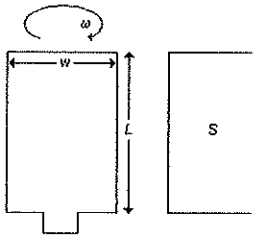
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14) A rectangular coil of wire (N = 20) that can rotate about an axis through its center is placed between the poles of a magnet in a magnetic field with a strength of 0.40 T, as shown in the figure. The length of the loop L is 0.16 m and its width w is 0.040 m. What is the rms voltage created by the generator if it is rotated at 200 Hz?

$A = LW = 0.0064\ m^2$
$N = 20$ loops of wire
$|\vec{B}| = 0.4\ T$
$f = 200\ Hz$

$V_{rms} = \frac{NAB2\pi f}{\sqrt{2}}$

$V_{rms} = 45.5\ V$



15) The coil in a 60 Hz ac generator has 125 turns, each having an area of 0.03 m^2 and is rotated in a uniform 0.12 T magnetic field. What is the peak output voltage of this generator?

$f = 60\,Hz$
$N = 125\ loops$
$A = 0.03\ m^2$

$V_p = NAB2\pi f = 170\,V$

16) The coil in a 60 Hz ac generator has 125 turns, each having an area of 0.03 m^2 and is rotated in a uniform magnetic field. If the rms voltage generated is 90 V, what is the magnetic field strength inside the generator?

$V_{rms} = \frac{V_p}{\sqrt{2}} = \frac{NAB2\pi f}{\sqrt{2}}$ Solve for B

$|\vec{B}| = \frac{V_{rms}(\sqrt{2})}{NA2\pi f} = 0.09\,T$

17) What is the main reason electric companies transmit the electrical energy at very high voltage?

A) It takes more energy to produce high current.
B) There is less resistance for high voltage.
C) High currents cost more to produce.
D) There is less energy loss when transmitting high voltage.

18) What is a typical voltage for high power transmission power lines?

A) 220 V
B) 12,000 V
C) 400,000 V
D) 120 V
E) 500 V

19) How many electrical grids make up the total grid in the continental United States?

A) 1
B) 2
C) 3
D) 4
E) 5

20) What is one advantage of the USA household power of 120 V over the European's 230 V

A) USA's 120 V has less loss of electrical energy.
B) USA's 120 V has a higher current.
C) USA's 120 V is safter.
D) There is no advantage.

KEY

1) A
2) C
3) A
4) B
5) 112 V
6) 15 A
7) 1,600 A
8) 400 A
9) 2 A
10) D
11) B
12) B
13) A
14) 45.5 V
15) 170 V
16) 0.09 T
17) D
18) C
19) C
20) C